

Lecture 20 - Interval Estimation
- Examples

Ex 1. $X_1, \dots, X_n$ iid observations from $N(\mu, \sigma^2)$.

To obtain $100(1-\alpha)\%$ C.I. for μ

Case (b) σ^2 unknown.

$\phi(T(X, \sigma), \mu)$ cannot be a pivot for here, $\because T(X, \sigma)$ involving unknown σ^2 is no longer a statistic.

Now $\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim N(0, 1)$, $\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$ are independent, so that,

$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$, can be used as a pivot

$$\text{Thus, } P\left[k_1 \leq \frac{\bar{X} - \mu}{S/\sqrt{n}} \leq k_2\right] = 1 - \alpha$$

$$\Leftrightarrow P\left[\bar{X} - k_2 \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} - k_1 \frac{S}{\sqrt{n}}\right] = 1 - \alpha$$

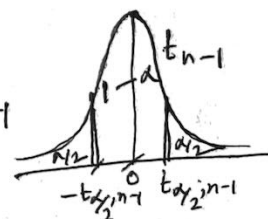
Here, we need to minimize the expected length, since the length $= \frac{S}{\sqrt{n}}(k_2 - k_1)$ involves a statistic (fnc of r.v.s $X_1, \dots, X_n$, and hence, itself a random qty) S .

$$\text{So, } L(k_1, k_2, \lambda) = \frac{E(S)}{\sqrt{n}}(k_2 - k_1) + \lambda \left(F_{t_{n-1}}(k_2) - F_{t_{n-1}}(k_1) - (1 - \alpha) \right)$$

Following the earlier argument we get $k_2 = -k_1$ (t_{n-1} distn is symmetric about 0, like the $N(0, 1)$)

$$\text{and hence } P\left[\bar{X} - k_2 \frac{S}{\sqrt{n}} \leq \mu \leq \bar{X} + k_2 \frac{S}{\sqrt{n}}\right] = 1 - \alpha$$

but here the pivot follows t_{n-1} distn, so that $k_2 = t_{\alpha/2, n-1}$



The shortest $100(1-\alpha)\%$ C.I. for μ is

$$\left[\bar{X} - t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}, \bar{X} + t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \right]$$

Ex 2 $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$, to obtain $100(1-\alpha)\%$ C.I. for σ^2

Case (a) μ known.

$$\frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \sim \chi^2_n$$

can be a pivot, $\because \mu$ is known

$$\text{so that, } P\left[k_1 \leq \frac{\sum_{i=1}^n (X_i - \mu)^2}{\sigma^2} \leq k_2\right] = 1 - \alpha$$

$$\text{Let } S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$$

$$\Leftrightarrow P\left[\frac{\sum_{i=1}^n (X_i - \mu)^2}{k_2} \leq \sigma^2 \leq \frac{\sum_{i=1}^n (X_i - \mu)^2}{k_1}\right] = 1 - \alpha$$

Here, length of the interval $= n S_n^2 \left(\frac{1}{k_1} - \frac{1}{k_2} \right)$ the constraint $F_{\chi_n^2}(k_2) - F_{\chi_n^2}(k_1) = 1 - \alpha$

To find the shortest ^{length} interval we consider

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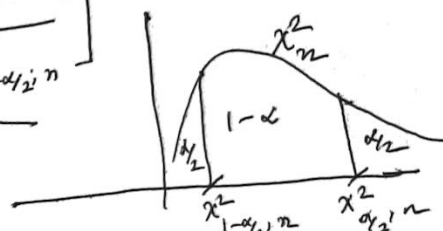
$$L(k_1, k_2, \lambda) = nE(S_n^2) \left(\frac{1}{k_1} - \frac{1}{k_2} \right) + \lambda (F_{\chi_n^2}(k_2) - F_{\chi_n^2}(k_1) - (1-\alpha))$$

$$\left. \begin{aligned} \frac{\partial L}{\partial k_1} &= -\frac{e}{k_1^2} - \lambda f_{\chi_n^2}(k_1) = 0 \\ \frac{\partial L}{\partial k_2} &= \frac{e}{k_2^2} + \lambda f_{\chi_n^2}(k_2) = 0 \end{aligned} \right\} \Rightarrow \frac{f_{\chi_n^2}(k_2)}{f_{\chi_n^2}(k_1)} = \frac{k_1^2}{k_2^2} \quad (*)$$

(it's χ_n^2 distn throughout)

Condition (*) may not necessarily lead to equal-tailed shortest length intervals. Hence, one convenient choice of $100(1-\alpha)\%$ C.I. is

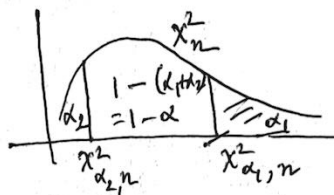
$$\left[\frac{nS_n^2}{\chi_{\alpha_2, n}^2}, \frac{nS_n^2}{\chi_{1-\alpha_2, n}^2} \right]$$



or, for any α_1, α_2 s.t. $\alpha_1 + \alpha_2 = \alpha$

$$\left[\frac{nS_n^2}{\chi_{\alpha_1, n}^2}, \frac{nS_n^2}{\chi_{\alpha_2, n}^2} \right]$$

is a $100(1-\alpha)\%$ C.I. for σ^2 (μ known)



Case (b) μ unknown

Here $\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} \sim \chi_{n-1}^2$ can be used as a pivot.

Denote $S = \frac{1}{n-1} \sum (x_i - \bar{x})^2$ & repeat the above argument to give $100(1-\alpha)\%$ C.I. for σ^2 (μ unknown) as

$$\left[\frac{(n-1)S^2}{\chi_{\alpha_2, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\alpha_2, n-1}^2} \right]$$

or, $\left[\frac{(n-1)S^2}{\chi_{\alpha_1, n-1}^2}, \frac{(n-1)S^2}{\chi_{\alpha_2, n-1}^2} \right]$ with $\alpha_1 + \alpha_2 = \alpha$
 $(\chi_{\alpha_1, n-1}^2 > \chi_{\alpha_2, n-1}^2)$